Smart Home Automation using Intelligent Agents:-

Introduction:-

A Smart home automation system uses Artificial Intelligence (AI) to control lighting, temperature, and Security based on user behaviour and Environmental Conditions. Intelligent agents Collect information through Sensors, make decisions and performs actions automatically. the system can also learn user preferences over time to improve Comfort, Energy efficiency, and Safty.

Simple Reflex Agent:-

A Simple reflex agent Selects actions based on the current input using condition-action rules.

* It reacts to present situation without considering past information
* Eg: If motion is detected, turn on the lights

model Based- agent :-

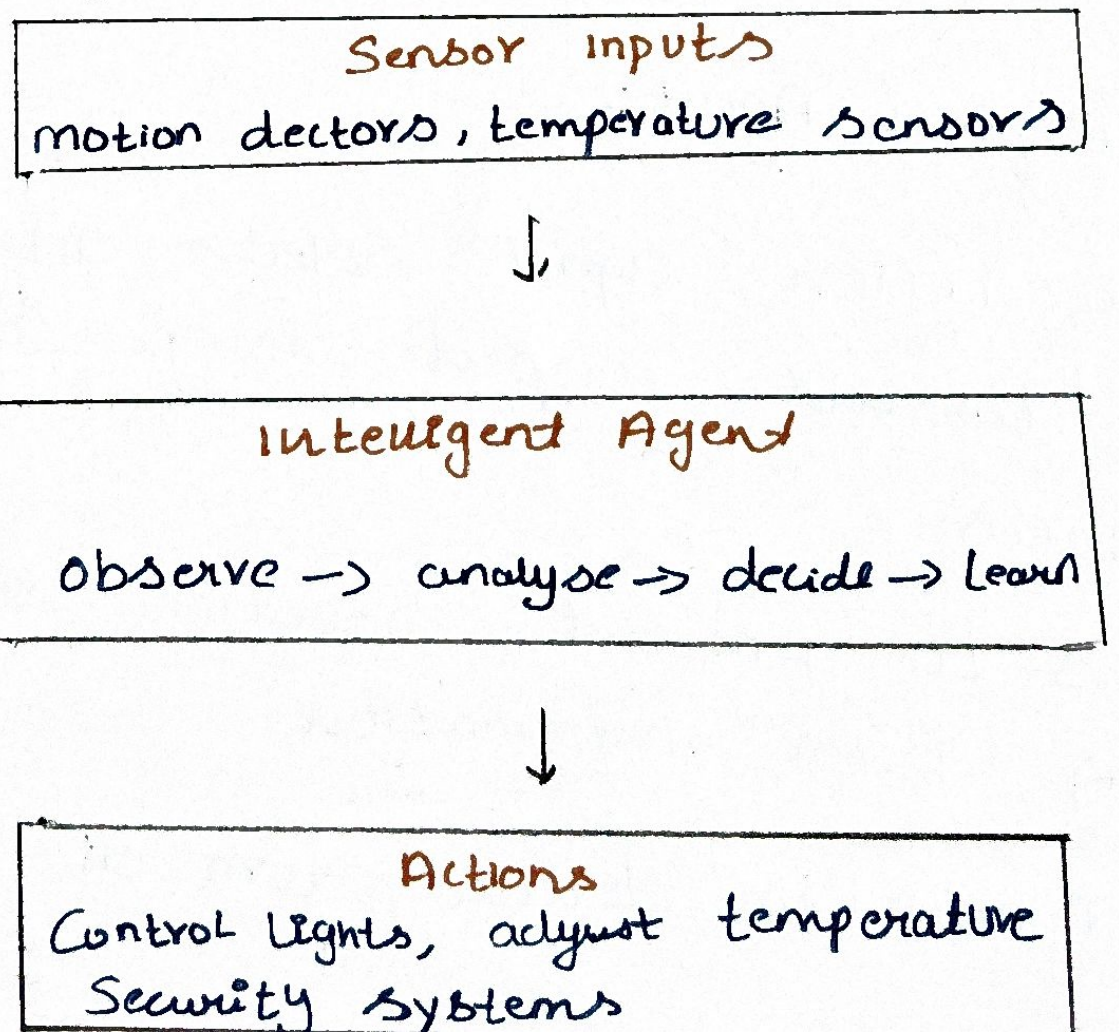
* remembers relevant information about the home's current state
* Eg: It consists room temperature whether a window is open and the air Conditioner's current status. Cooling

Goal Based Agent :-

It Evaluates possible actions according to the desired outcome.

Eg: It adjust thermostat to reach the user's perferred temperature

Working of Smart Home System



Best Approach:-

* The model based Component maintains informations about home's Environment
* The goal-based Component work toward target
* Simple reflex rules can handle urgent actions, such as activating alarm security sensors detects a threat

Conclusion:-

Intelligent agents make Smart homes more adaptive, efficient and secure. A hybrid approach provides fast responses support improving Comfort, Energy efficiency and home Security.

Robot Maze Navigation using UCS and IDS

Introduction

A robot placed in an unknown maze must find an Exit without having prior knowledge of the layout. It can use uninformed Search algorithms to Explore possible paths Systematically. Two important uninformed Search Strategies are Uniform Cost Search (UCS) and Iterative Deepening Search (IDS)

Uniform Cost Search (UCS)

* Start from the initial position
* Store the available paths in a priority queue
* Explore it's unvisited neighbouring positions
* Repeat until the Exit is reached

Advantages:-

- Finds least cost path when step costs are non-negative
- Suitable when different paths have different movement costs
- Complete understandard Conditions, such as a finite branching factor and a positive minimum step cost.

Iterative Deepening Search (IDS)

Iterative Deepening Search repeatedly performs depth limited Search, increasing depth limit

Advantages:-

- uses less memory than breadth first Search
- Compute for finite branching factors
- Finds a shallowest solution Every steps

Disadvantages

- Repeats Exploration of upper-level nodes
- may waste time revisiting same positions
- Does not guarantee the least-cost path

Recommended Combination:-

model - based, goal - based agent using UCS is appropriate when the maze has differed movement costs, such as dangerous paths or Energy - intensive routes. The robot can update its internal map as it Explores and update its least - cost path through known portion of the maze

Conclusion

UCS and IDS are useful uninformed Search algorithms for maze navigation. UCS is preferable when finding the minimum - cost path is important. Whereas IDS is preferable when memory is limited and all steps have Equal cost Combining a model based agent with goal ba Planning Enables the robot the Explore, learn And reach the Exit efficiently